

EMCH 367 Controls

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Fall 2024

Homework #4

Due date: November 3, 11:59 pm

Problem 1. (50pt) Consider the closed loop transfer function given by

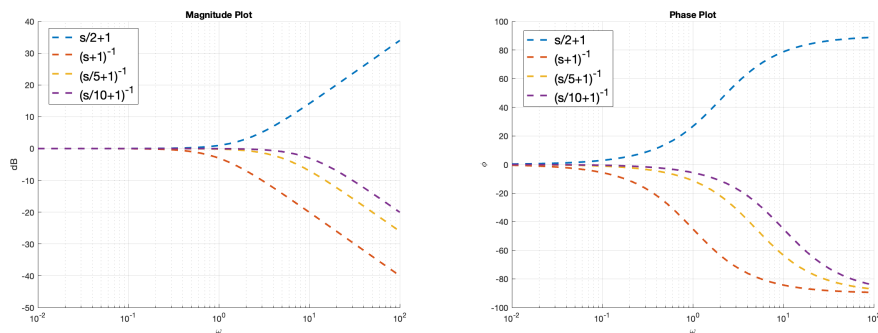
$$G_{cl}(s) = \frac{25(s+2)}{(s+1)(s+5)(s+10)}.$$

(a) (20pt) First, write the closed loop transfer function as a standard form. Then draw the bode diagram for each element from $\omega = 0.01$ rad/s to $\omega = 100$ rad/s.

Solution: First, note that

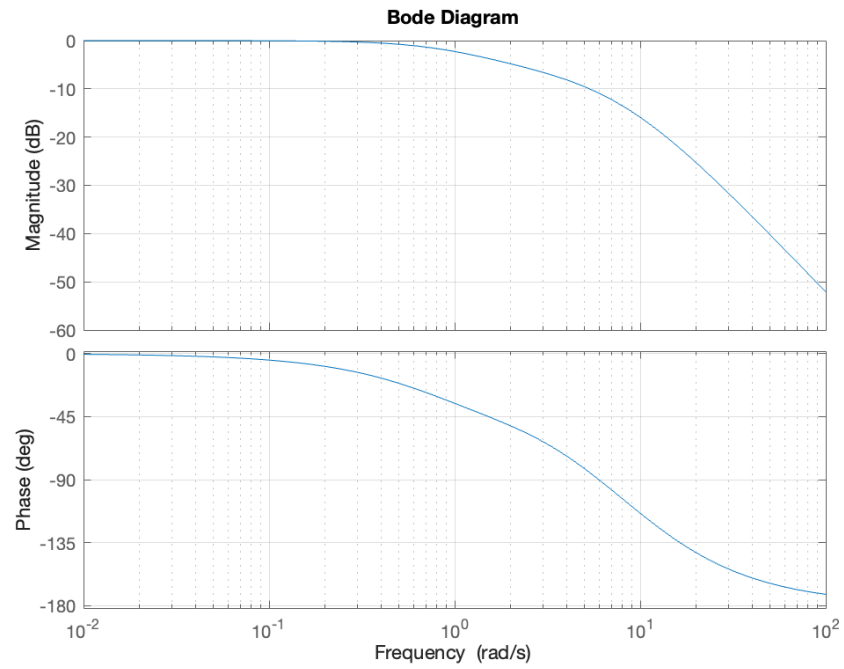
$$\begin{aligned} G_{cl}(s) &= \frac{25 \cdot 2 \left(\frac{s}{2} + 1 \right)}{(s+1) \cdot 5 \left(\frac{s}{5} + 1 \right) \cdot 10 \left(\frac{s}{10} + 1 \right)} \\ &= \frac{\left(\frac{s}{2} + 1 \right)}{(s+1) \left(\frac{s}{5} + 1 \right) \left(\frac{s}{10} + 1 \right)} \end{aligned}$$

The bode diagram for each element is given as follows.



(b) (10pt) Draw the Bode diagram of $G_{cl}(s)$ using MATLAB.

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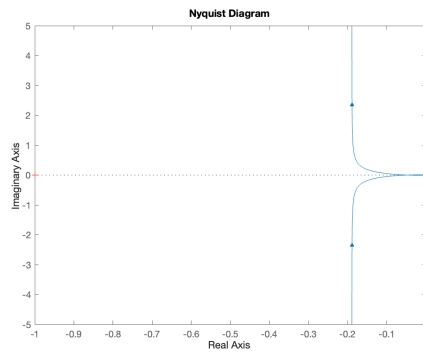


(c) (10pt) Draw the Nyquist plot using MATLAB. (hint: First, find the open loop transfer function)

Solution: First, note that

$$\begin{aligned}
 G_{cl}(s) &= \frac{25(s+2)}{(s+1)(s+5)(s+10)} \\
 &= \frac{25s+50}{s^3+16s^2+65s+50} \\
 &= \frac{25(s+2)}{s^3+16s^2+65s} \\
 &= \frac{1 + \frac{50}{s^3+16s^2+65s}}{1 + \frac{25(s+2)}{s(s^2+16s+65)}} \\
 &= \frac{s(s^2+16s+65)}{1 + \frac{50}{s(s^2+16s+65)}}
 \end{aligned}$$

The Nyquist plot of $s(s^2+16s+65)$ is given by



(d) (10pt) Analyze the stability of the closed loop system using Nyquist stability criterion.

Solution: Note that the open loop transfer function is given by

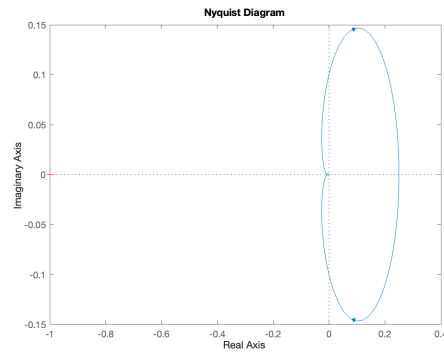
$$\frac{50}{s(s^2 + 16s + 65)},$$

where the pole is located at the origin and LHP. Therefore, we have $P = 0$. Additionally, it follows from (c) that $N = 0$, and hence, $Z = 0$. Therefore, the system is stable.

You can use other forms of open loop transfer functions as well. For example, we will use different open loop transfer function

$$\begin{aligned} G_{cl}(s) &= \frac{25s + 50}{s^3 + 16s^2 + 65s + 50} \\ &= \frac{25(s + 2)}{s^3 + 16s^2 + 65s + 40 + 10} \\ &= \frac{25(s + 2)}{s^3 + 16s^2 + 65s + 40} \\ &= \frac{1}{1 + \frac{10}{s^3 + 16s^2 + 65s + 40}}, \end{aligned}$$

and the Nyquist plot is given by the following.



Problem 2. (30pt)

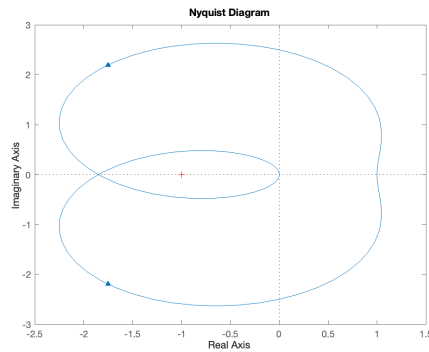
(a) (15pt) Consider a unity feedback control system with the following open-loop transfer function.

$$G(s) = \frac{s^2 + 2s + 1}{s^3 + 0.2s^2 + s + 1}.$$

Draw a Nyquist plot of $G(s)$ using MATLAB and examine the stability of the closed-loop system.

Solution: The poles are located at $p_1 = -0.72458$, $p_2 = 0.26229 + 1.14512j$, and $p_3 = 0.26229 - 1.14512j$. Therefore, $P = 2$.

The Nyquist plot for $G(s)$ is given by



Note that $N = -2$. Thus, it follows from Nyquist stability criterion that $Z = N + P = 0$, and hence, the system is stable.

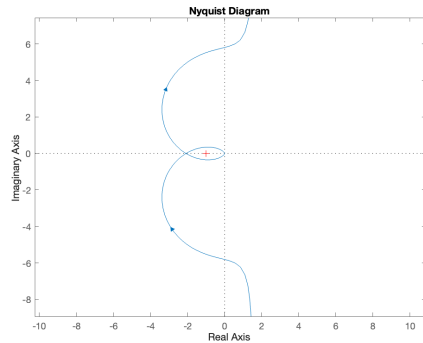
(b) (15pt) Consider a unity feedback control system with the following open-loop transfer function.

$$G(s) = \frac{100(s+1)}{s(s+5)(s^2+2s+5)}.$$

Draw a Nyquist plot of $G(s)$ using MATLAB and examine the stability of the closed-loop system.

Solution: The poles are located at $p_1 = 0$, $p_2 = -5$, $p_3 = -1 + 2j$, and $p_4 = -1 - 2j$. Therefore, $P = 0$.

The Nyquist plot for $G(s)$ is given by



Note that $N = 2$. Thus, it follows from Nyquist stability criterion that $Z = N + P = 2$, and hence, the system is unstable.

Problem 3. (20pt) Consider the linearized modeling of an inverted pendulum on a cart.

$$\begin{aligned}(M + m)\ddot{x}(t) + b\dot{x}(t) &= f(t) + ml\ddot{\theta}(t), \\ (J + ml^2)\ddot{\theta}(t) - mgl\theta(t) &= ml\ddot{x}(t).\end{aligned}$$

(a) (10pt) Let the state variable be $z = [x, \dot{x}, \theta, \dot{\theta}]^T$. Write the equation of motion in state-space form and identify A and B matrices.

Solution: First, note that it follows from the second equation that

$$\ddot{\theta} = \frac{mgl}{J + ml^2}\theta + \frac{ml}{J + ml^2}\ddot{x}.$$

Substituting this into the first equation will yield

$$\begin{aligned}(M + m)\ddot{x} + b\dot{x} &= f + \frac{m^2gl^2}{J + ml^2}\theta + \frac{m^2l^2}{J + ml^2}\ddot{x} \\ \implies \ddot{x} &= \left[M + m - \frac{m^2l^2}{J + ml^2} \right]^{-1} \left[-b\dot{x} + \frac{m^2gl^2}{J + ml^2}\theta + f \right].\end{aligned}$$

Next, it follows from the first equation that

$$\ddot{x} = \frac{-b}{M + m}\dot{x} + \frac{ml}{M + m}\ddot{\theta} + \frac{1}{M + m}f$$

Substituting this into the second equation yields

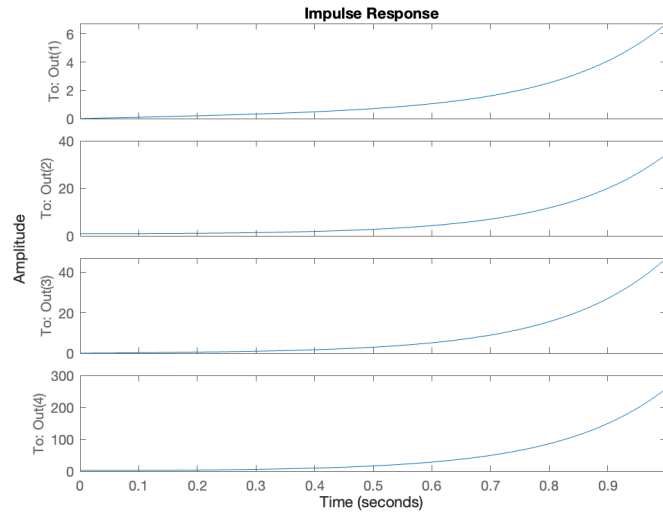
$$\begin{aligned}\ddot{\theta} \left[J + ml^2 - \frac{m^2l^2}{M + m} \right] &= -\frac{mlb}{M + m}\dot{x} + mgl\theta + \frac{ml}{M + m}f \\ \implies \ddot{\theta} &= \left[J + ml^2 - \frac{m^2l^2}{M + m} \right]^{-1} \left[-\frac{mlb}{M + m}\dot{x} + mgl\theta + \frac{ml}{M + m}f \right].\end{aligned}$$

Now, it follows that

$$\begin{aligned}
 \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{\theta} \\ \ddot{\theta} \end{bmatrix} &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\left[M + m - \frac{m^2 l^2}{J + ml^2}\right]^{-1} b & \left[M + m - \frac{m^2 l^2}{J + ml^2}\right]^{-1} \frac{m^2 g l^2}{J + ml^2} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -\left[J + ml^2 - \frac{m^2 l^2}{M + m}\right]^{-1} \frac{mlb}{M + m} & \left[J + ml^2 - \frac{m^2 l^2}{M + m}\right]^{-1} mgl & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} \\
 &+ \begin{bmatrix} 0 \\ \left[M + m - \frac{m^2 l^2}{J + ml^2}\right]^{-1} \\ 0 \\ \left[J + ml^2 - \frac{m^2 l^2}{M + m}\right]^{-1} \frac{ml}{M + m} \end{bmatrix} f \\
 &= Az + Bf
 \end{aligned}$$

(b) (10pt) Let $M = 1$ kg, $m = 0.5$ kg, $b = 0.2$ N/(m · s), $l = 0.4$ m, $J = 0.01$ k · m². Using MATLAB, plot impulse response using state space model for 1 second.

Solution:



Bonus Problem. (20pt) Consider the suspension system given in Figure 1. Let the state vector be given by $z(t) = [x(t), \dot{x}(t), y(t), \dot{y}(t)]$. Find A and B of the following state space equation.

$$\dot{z}(t) = Az(t) + Bu(t).$$

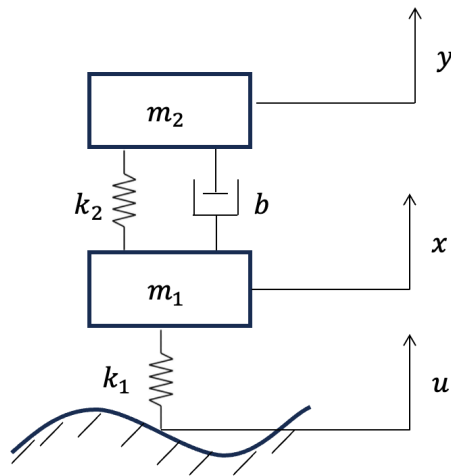
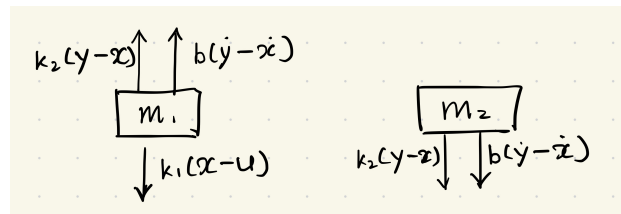


Figure 1

Solution: The free body diagram is given by



and hence, the equation of motion is given by

$$\begin{aligned} m_1 \ddot{x} &= k_2(y - x) + b(\dot{y} - \dot{x}) - k_1(x - u), \\ m_2 \ddot{y} &= -k_2(y - x) - b(\dot{y} - \dot{x}). \end{aligned}$$

Therefore, it follows that

$$\dot{z}(t) = \begin{bmatrix} \dot{x}(t) \\ \ddot{x}(t) \\ \dot{y}(t) \\ \ddot{y}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_1+k_2}{m_1} & -\frac{b}{m_1} & \frac{k_2}{m_1} & \frac{b}{m_1} \\ 0 & 0 & 0 & 1 \\ \frac{k_2}{m_2} & \frac{b}{m_2} & -\frac{k_2}{m_2} & -\frac{b}{m_2} \end{bmatrix} \begin{bmatrix} x(t) \\ \dot{x}(t) \\ y(t) \\ \dot{y}(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{k_1}{m_1} \\ 0 \\ 0 \end{bmatrix} u(t).$$